

METLAKATLA, AK, USA

WMO: 703985

Lat: 55.131N

Lon: 131.578W

Elev: 0

StdP: 14.70

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 25377

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.0	26.6	-8.9	3.4	35.6	0.6	5.7	34.0	26.1	40.2	24.0	40.3	10.5	80	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	69.9	59.9	66.0	58.8	63.8	56.9	61.1	67.6	59.6	64.5	58.4	63.0	6.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	70.5	62.6	56.9	68.9	62.0	55.4	65.2	60.4	27.0	67.6	26.0	65.0	25.2	62.8	68.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.0	17.9	17.6	74.5	6.0	5.0	13.3	78.1	9.8	81.0	6.5	83.8	2.1	87.5
			13.9	62.9	5.1	2.4	10.3	64.6	7.3	66.0	4.4	67.3	0.7	69.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	38.9	38.8	39.6	43.6	49.3	54.4	58.0	58.4	54.0	47.6	41.8	38.8
	DBStd	8.73	6.53	5.04	5.16	4.03	4.46	4.01	3.66	3.28	3.03	4.15	4.92	5.87
	HDD50	1942	346	315	322	196	67	3	0	0	3	93	248	349
	HDD65	6581	811	734	787	641	487	319	220	207	329	538	697	811
	CDD50	847	1	1	1	5	45	136	249	261	124	20	1	3
	CDD65	9	0	0	0	0	0	2	4	3	0	0	0	0
	CDH74	16	0	0	0	0	1	9	4	2	0	0	0	0
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	7.5	9.5	8.3	8.5	7.2	6.7	6.3	5.9	5.6	6.2	7.9	9.0	9.5
Precipitation	PrecAvg	118.9	12.5	9.3	9.4	8.1	6.4	5.3	5.7	8.4	12.1	15.2	13.7	12.7
	PrecMax	145.6	20.3	18.5	17.2	17.2	16.3	8.8	11.6	16.6	21.2	26.3	23.3	24.1
	PrecMin	85.5	3.9	3.8	4.2	2.0	0.3	2.2	0.8	3.0	4.6	5.6	8.2	3.9
	PrecStd	15.6	3.8	3.8	3.2	3.7	3.8	2.0	2.6	4.0	4.0	5.0	3.5	4.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	53.9	54.3	61.1	69.6	73.1	73.0	72.8	65.2	60.6	54.1	61.3
		MCWB	44.0	43.6	45.8	47.4	53.3	59.7	62.2	61.3	57.5	50.9	49.0	42.2
	2%	DB	50.0	49.8	50.2	54.6	62.8	66.3	69.8	68.5	62.7	56.6	51.7	51.6
		MCWB	45.7	43.6	43.8	46.2	50.3	57.7	61.0	60.0	56.4	51.3	48.9	45.2
	5%	DB	47.8	46.4	47.7	52.0	57.4	63.3	65.8	65.6	60.6	54.5	49.7	48.0
		MCWB	44.6	42.1	43.1	45.6	48.6	55.7	59.4	59.5	55.2	50.3	47.1	44.0
	10%	DB	46.0	45.4	45.8	48.3	55.1	60.8	63.4	63.4	58.0	53.7	47.9	45.9
		MCWB	42.9	41.6	41.8	43.7	48.1	54.5	57.3	57.7	53.7	49.8	45.3	42.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	48.0	47.9	50.7	55.4	61.0	63.3	62.6	59.9	54.4	51.2	47.3
		MCDB	52.0	50.6	52.2	57.5	64.0	71.6	70.9	68.7	63.6	57.1	52.2	49.6
	2%	WB	46.4	45.3	45.7	47.8	52.7	58.3	61.4	61.0	57.8	52.8	49.3	46.1
		MCDB	48.2	46.6	47.9	52.5	59.2	65.3	68.1	66.9	61.3	54.8	51.3	48.0
	5%	WB	45.4	43.4	44.1	46.1	51.1	56.7	59.6	59.6	56.3	51.3	47.0	45.1
		MCDB	46.8	46.5	46.8	49.3	56.2	62.6	64.7	64.2	59.4	53.5	48.9	47.1
	10%	WB	44.0	41.8	42.3	44.8	49.9	54.9	58.2	58.6	54.9	50.1	45.7	43.4
		MCDB	46.1	45.5	45.8	47.8	54.5	59.9	62.9	62.9	57.6	52.5	47.2	47.1
Mean Daily Temperature Range	5% DB	MDBR	6.4	6.6	7.3	8.4	8.7	8.4	7.8	7.7	7.1	6.6	6.1	6.0
		MCDBR	8.1	8.2	10.3	12.4	13.1	14.1	12.9	12.4	9.7	8.0	7.1	8.4
		MCWBR	5.8	5.5	6.7	7.0	7.1	7.6	6.9	6.7	5.8	5.5	6.1	5.3
	5% WB	MCDBR	6.4	7.2	8.9	10.2	11.2	12.9	11.4	10.7	8.5	6.7	6.7	6.7
		MCWBR	5.9	5.7	6.9	6.7	6.5	7.3	6.6	6.3	5.7	5.5	6.4	5.8
Clear-Sky Solar Irradiance	taub	0.300	0.326	0.335	0.366	0.364	0.368	0.375	0.360	0.345	0.336	0.309	0.294	
	taud	2.380	2.386	2.438	2.358	2.436	2.457	2.472	2.541	2.560	2.478	2.350	2.280	
	Ebn at Noon	204	239	267	272	279	279	274	272	262	234	195	173	
	Edh at Noon	16	23	27	34	33	33	32	28	24	21	16	14	
All-Sky Solar Radiation	RadAvg	151	401	710	1169	1472	1534	1369	1197	780	427	192	118	
	RadStd	20	80	117	90	151	119	175	117	103	69	32	22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page